

Agile Sprint Planning & User Story Workshop

Project: Intelligent E-Governance Citizen Service Request Management System

1. Introduction

Project Background

The Intelligent E-Governance Citizen Service Request Management System is designed to improve the way government departments receive, process, track, and resolve citizen service requests through the use of modern digital technologies. Traditional government service management often depends on manual paperwork, physical office visits, and disconnected departmental processes. These conventional methods result in long waiting times, delayed responses, lack of transparency, poor communication, and difficulty in tracking the status of requests. As citizen expectations continue to grow in the digital era, governments are under increasing pressure to provide services that are faster, more transparent, and easily accessible.

E-governance has emerged as a powerful approach to transforming public administration by integrating information and communication technologies into government operations. It enables citizens to access public services online without the need for repeated visits to government offices. However, many existing e-governance systems still face challenges such as inefficient request routing, delayed grievance resolution, lack of real-time updates, and limited decision support for government officials. These issues reduce citizen satisfaction and affect the overall efficiency of public service delivery.

The Intelligent E-Governance Citizen Service Request Management System addresses these challenges by combining cloud computing, workflow automation, artificial intelligence (AI), and data analytics into a unified platform. Citizens can submit service requests or complaints through a web or mobile application, upload supporting documents, and monitor the progress of their requests in real time. The system automatically categorizes requests based on their type, assigns them to the appropriate government department, prioritizes urgent issues, and sends notifications to both citizens and officials throughout the resolution process.

Artificial Intelligence plays a significant role in improving the efficiency of the system. AI techniques such as Natural Language Processing (NLP) can analyse citizen complaints, identify the nature of the issue, and classify requests automatically. Machine learning algorithms help predict processing times, identify high-priority cases, and recommend suitable actions to government staff. Workflow automation reduces manual intervention, minimizes human errors, and ensures that service requests follow predefined procedures, resulting in faster response times.

The system also enhances transparency and accountability by maintaining complete digital records of every request, including submission details, processing history, assigned officials, and completion status. Citizens receive regular updates through email, SMS, or mobile notifications, increasing trust in government services. Administrators can monitor departmental performance using dashboards and analytical reports that display key performance indicators such as request volumes, average resolution time, pending cases, and citizen satisfaction levels.

Another important aspect of the project is data security and privacy. Since government systems handle sensitive personal information, the platform incorporates secure authentication, role-based access control, encrypted communication, and secure cloud storage to protect user data. These security measures ensure compliance with digital governance standards while safeguarding citizen information from unauthorized access.

Problem Statement

Despite the increasing adoption of digital governance platforms, many government agencies still face challenges in delivering public services that are transparent, efficient, and timely. Traditional manual workflows and fragmented systems lead to delays, lack of accountability, poor tracking of citizen requests, and inconsistent service quality. Citizens expect real-time updates and quick resolutions, but existing infrastructure often struggles to scale, integrate data, and automate decision-making. Without leveraging technologies like AI, cloud computing, and workflow automation, governments find it difficult to optimize resource allocation, reduce administrative overhead, and enhance citizen satisfaction. There is a need for an intelligent, cloud-based digital governance platform that automates service request management, ensures transparency, and provides efficient, data-driven responses to improve public service delivery and administrative efficiency.

Project Vision

To build an AI-powered smart aquaculture platform that continuously monitors fish farms, predicts disease risks, optimizes feeding, and supports sustainable fish farming through IoT-enabled automation and cloud-based remote monitoring.

Project Mission

To provide fish farmers with an intelligent decision-support system that: Improves fish survival rates
Increases overall productivity Reduces operational costs Minimizes environmental impact Enables
real-time remote farm management

Objectives

- Monitor water quality parameters
- continuously Predict fish
- diseases using AI/ML models Track
- fish growth and health patterns
- Optimize feeding schedules to reduce wastage
- Generate real-time environmental
- alerts Reduce fish mortality
- Improve production yield Lower
- operational expenses
- Enable secure cloud-based remote monitoring and control

Expected Outcomes

- Healthy and disease-free fish population
- Higher production and yield
- Improved water quality management
- Reduced feed wastage Early
- detection of diseases and risks
- Lower operational and maintenance costs
- Promotion of sustainable
- aquaculture practice
- Increased profitability for fish farmers

Technologies Used

Technology	Purpose
IoT Sensors	Collect water quality data
Artificial Intelligence	Disease prediction
Machine Learning	Feeding optimization
Cloud Computing	Data storage
Mobile App	Remote monitoring
Web Dashboard	Farm management
Data Analytics	Reports & trends
GPS	Farm location tracking
SMS Gateway	Emergency alerts

Quality Parameters

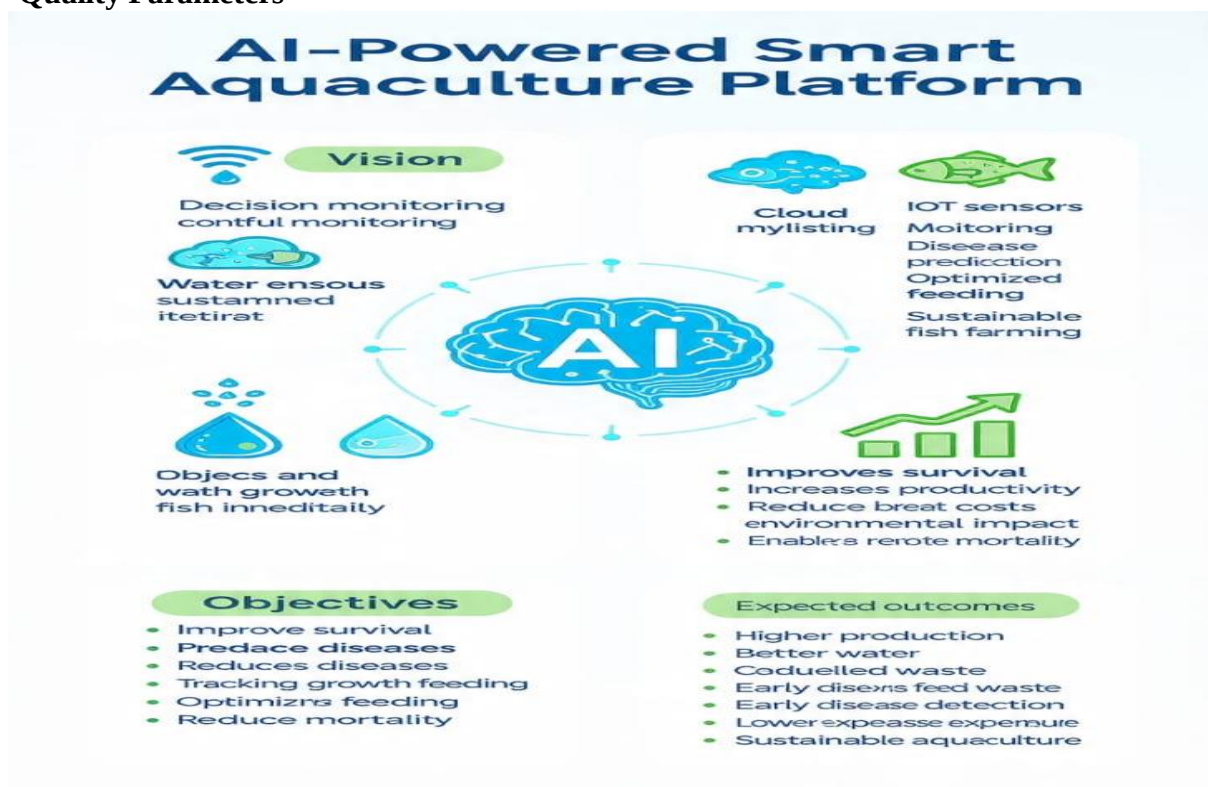


Table 6

Parameter	Ideal Range	Importance
Temperature	24–30°C	Fish metabolism
pH	6.5–8.5	Healthy environment
Dissolved Oxygen	>5 mg/L	Fish survival
Ammonia	<0.02 mg/L	Prevent toxicity
Turbidity	Low	Water clarity
Salinity	Species dependent	Growth

Major Components

Hardware

- Temperature Sensor
- pH Sensor
- Dissolved Oxygen Sensor
- Ammonia Sensor
- Turbidity Sensor
- ESP32/Arduino
- Wi-Fi Module
- Solar Power (Optional)

Software

- React / Flutter
- Python
- TensorFlow
- Node.js
- Firebase
- MySQL
- AWS / Azure Cloud

Stakeholders

Stakeholder	Responsibility
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Fish Farmer	Uses system daily
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Farm Manager	Monitors production
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Veterinarian	Disease diagnosis
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AI Engineer	Develops ML model
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IoT Engineer	Sensor maintenance
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Government	Regulatory monitoring
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Investors	Business performance
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Customers	Product quality
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Functional Requirements

Water Monitoring

- Read sensor values every minute
- Store historical data
- Display graphs
- Detect abnormal values

Disease Prediction

- Analyse water quality
- Analyse fish behaviour
- Predict diseases
- Suggest treatment

Feeding Management

- Recommend feeding quantity
- Recommend feeding time
- Record feeding history
- Reduce feed waste

Fish Growth

- Record fish weight
- Record fish length

- Calculate growth rate
- Compare production

Alert System

- High ammonia alert
- Low oxygen alert
- High temperature alert
- Disease risk alert
- Mobile notification
- SMS notification

Non-Functional Requirements

- Secure login
- Cloud backup
- Fast response (<3 seconds)
- 99% uptime
- User-friendly interface
- Mobile compatibility
- Data encryption
- Scalable architecture

Agile Methodology

The project follows **Scrum**.

Sprint Duration:

2 Weeks

Scrum Events

- Sprint Planning
- Daily Stand-up
- Sprint Review
- Sprint Retrospective

Product Roadmap

Sprint Module

Sprint 1 Authentication + Dashboard

Sprint 2 Water Monitoring

Sprint 3 Alert System

Sprint 4 AI Disease Prediction

Sprint 5 Feeding Optimization

Sprint 6 Fish Growth Analytics

Sprint 7 Reports

Sprint 8 Final Testing

Epic Breakdown

Epic 1

Water Quality Monitoring

Features

- Sensor Integration
- Dashboard
- Graphs
- Historical Data

Epic 2

Disease Prediction

Features

- AI Model
- Prediction
- Disease History
- Recommendations

Epic 3

Feeding Optimization

Features

- Feeding Calculator
- Schedule

- Feed History

Epic 4

Growth Monitoring

Features

- Weight Records
- Growth Graph
- Production Report

Epic 5

Alerts

Features

- SMS
- Email
- Mobile Notification
- Emergency Warning

User Stories

Epic 1

US-01

As a Fish Farmer,

I want to monitor water quality,

So that I can maintain healthy fish.

US-02

As a Farm Manager,

I want historical graphs,

So that I can analyse trends.

Epic 2

US-03

As a Fish Farmer,

I want disease prediction,

So that I can prevent outbreaks.

US-04

As a Veterinarian,

I want AI-generated disease reports,

So that I can recommend treatment.

Epic 3

US-05

As a Fish Farmer,

I want optimised feeding schedules,

So that feed wastage is minimised.

Epic 4

US-06

As a Farm Manager,

I want growth analytics,

So that production can be evaluated.

Epic 5

US-07

As a Fish Farmer,

I want instant notifications,

So that I can respond quickly.

Acceptance Criteria

US-01

Given

Sensors are active

When

The farmer opens the dashboard

Then

Live water quality should be displayed.

US-03

Given

Environmental data is available

When

AI processes the data

Then

Disease prediction should be generated.

US-05

Given

Growth records exist

When

The system analyses feed history

Then

Recommended feeding schedule should be displayed.

US-07**Given**

Oxygen level falls below threshold

When

The event occurs

Then

SMS and mobile notification should be sent immediately.

Product Backlog

Priority	Feature	Story Points
High	Login	3
High	Water Monitoring	8
High	Dashboard	5
High	Alerts	5
Medium	Disease Prediction	13
Medium	Feeding Optimization	8
Medium	Growth Tracking	8
Low	Reports	5
Low	Analytics	8

Sprint 1 Planning

Sprint Goal

Develop the core platform with authentication, sensor integration, dashboard, and environmental alerts.

Sprint Backlog

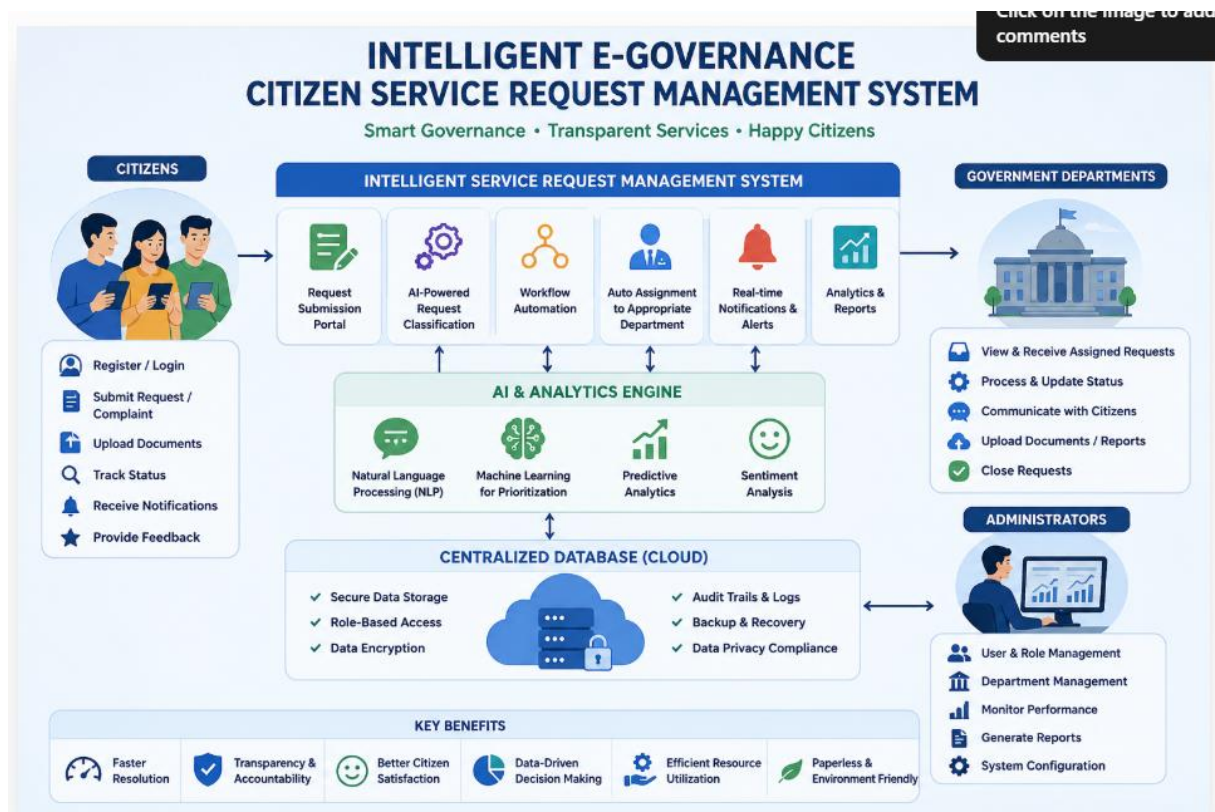
Story	Tasks
Login	UI, Authentication
Dashboard	Charts, Live Data
Sensors	ESP32 Integration
Alerts	SMS API
Database	Store Sensor Data

Story Point Estimation

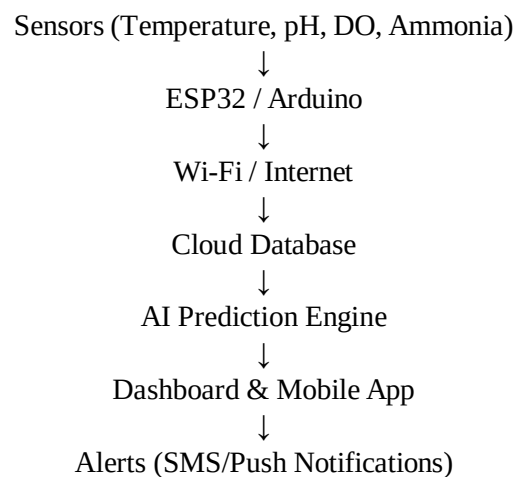
Story	Points	Reason
Login	3	Simple
Dashboard	5	Medium UI
Sensor Integration	8	Hardware integration
Alerts	5	Notification APIs
Disease Prediction	13	AI complexity
Feeding Module	8	Business logic
Growth Module	8	Analytics

Estimation Technique: Fibonacci (1, 2, 3, 5, 8, 13)

System Architecture



Flow:



Risk Analysis

Risk	Mitigation
Sensor failure	Backup sensors and periodic calibration
Internet outage	Local data buffering and sync
AI prediction errors	Retrain models with updated datasets
Power failure	UPS or solar backup
High operational cost	Optimise cloud resource usage

Future Enhancements

- Drone-based pond surveillance
- Underwater cameras for fish behaviour analysis
- Computer vision for automatic fish counting
- Automatic feeder integration
- Weather forecasting integration
- Blockchain-based seafood traceability
- Digital twin of fish ponds
- Predictive maintenance for farm equipment

Conclusion

E-governance has emerged as a powerful approach to transforming public administration by integrating information and communication technologies into government operations. It enables citizens to access public services online without the need for repeated visits to government offices. However, many existing e-governance systems still face challenges such as inefficient request routing, delayed grievance resolution, lack of real-time updates, and limited decision support for government officials. These issues reduce citizen satisfaction and affect the overall efficiency of public service delivery.